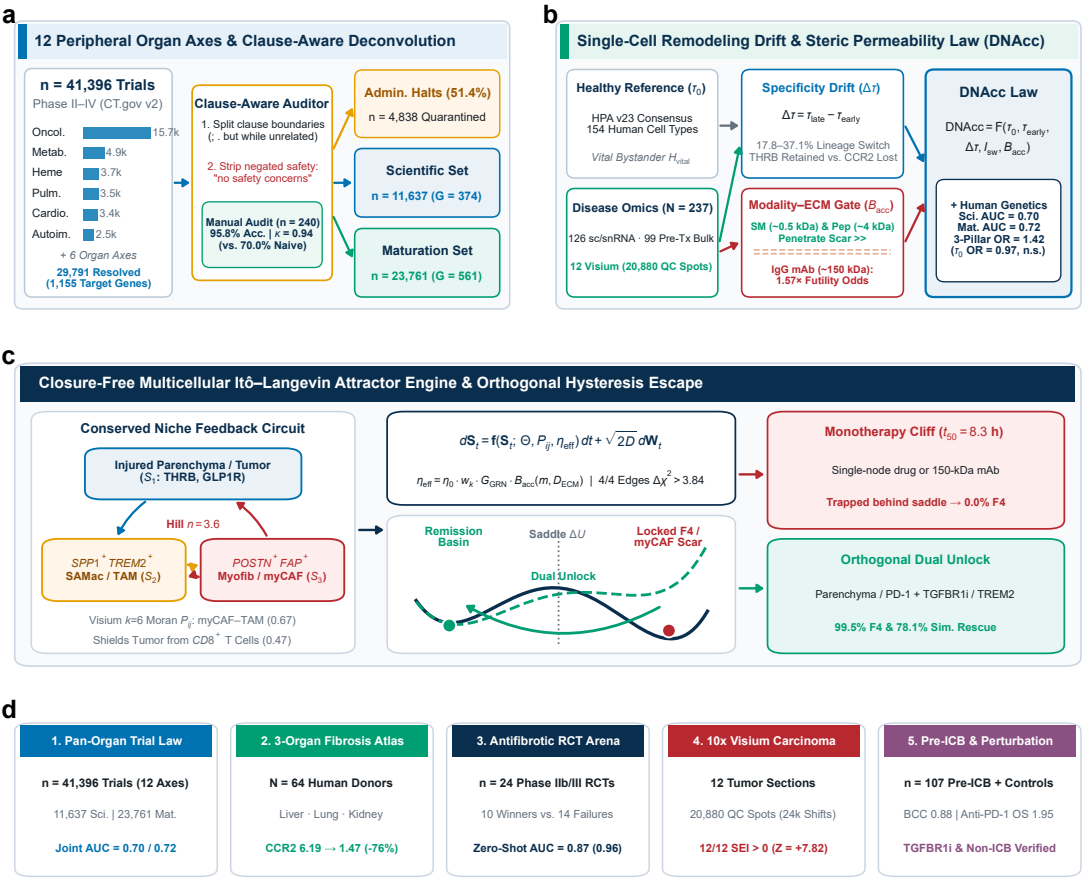

a
12 Peripheral Organ Axes & Clause-Aware Deconvolution
n = 41,396 Trials
Phase II–IV (CT.gov v2)
Oncol. 15.7k
Metab. 4.9k
Heme 3.7k
Pulm. 3.5k
Cardio. 3.4k
Autoim. 2.5k
+ 6 Organ Axes
29,791 Resolved
(1,155 Target Genes)
Clause-Aware Auditor
1. Split clause boundaries
(; . but while unrelated)
2. Strip negated safety:
"no safety concerns"
Manual Audit (n = 240)
95.8% Acc. | κ = 0.94
(vs. 70.0% Naive)
Admin. Halts (51.4%)
n = 4,838 Quarantined
Scientific Set
n = 11,637 (G = 374)
Maturation Set
n = 23,761 (G = 561)
b
Single-Cell Remodeling Drift & Steric Permeability Law (DNAcc)
Healthy Reference (τ_0)
HPA v23 Consensus
154 Human Cell Types
Vital Bystander H_{vital}
Disease Omics (N = 237)
126 sc/snRNA · 99 Pre-Tx Bulk
12 Visium (20,880 QC Spots)
Specificity Drift ($\Delta\tau$)
$\Delta\tau = \tau_{late} - \tau_{early}$
17.8–37.1% Lineage Switch
THRB Retained vs. CCR2 Lost
Modality–ECM Gate (B_{acc})
SM (~0.5 kDa) & Pep (~4 kDa)
Penetrate Scar >>
IgG mAb (~150 kDa):
1.57× Futility Odds
DNAcc Law
$DNAcc = F(\tau_0, \tau_{early}, \Delta\tau, I_{sw}, B_{acc})$
+ Human Genetics
Sci. AUC = 0.70
Mat. AUC = 0.72
3-Pillar OR = 1.42
(τ_0 OR = 0.97, n.s.)
c
Closure-Free Multicellular Itô–Langevin Attractor Engine & Orthogonal Hysteresis Escape
Conserved Niche Feedback Circuit
Injured Parenchyma / Tumor
(S_1: THRB, GLP1R)
Hill $n = 3.6$
$SPP1^+ TREM2^+$
SAMac / TAM (S_2)
$POSTN^+ FAP^+$
Myofib / myCAF (S_3)
Visium k=6 Moran P_{ij}: myCAF–TAM (0.67)
Shields Tumor from $CD8^+$ T Cells (0.47)
$d\mathbf{S}_t = \mathbf{f}(\mathbf{S}_t; \Theta, P_{ij}, \eta_{eff})\,dt + \sqrt{2D}\,d\mathbf{W}_t$
$\eta_{eff} = \eta_0 \cdot w_k \cdot G_{GRN} \cdot B_{acc}(m, D_{ECM})$ | 4/4 Edges $\Delta\chi^2 > 3.84$
Remission Basin
Saddle ΔU
Locked F4 / myCAF Scar
Dual Unlock
Monotherapy Cliff ($t_{50} = 8.3$ h)
Single-node drug or 150-kDa mAb
Trapped behind saddle → 0.0% F4
Orthogonal Dual Unlock
Parenchyma / PD-1 + TGFBR1i / TREM2
99.5% F4 & 78.1% Sim. Rescue
d
1. Pan-Organ Trial Law
n = 41,396 Trials (12 Axes)
11,637 Sci. | 23,761 Mat.
Joint AUC = 0.70 / 0.72
2. 3-Organ Fibrosis Atlas
N = 64 Human Donors
Liver · Lung · Kidney
CCR2 6.19 → 1.47 (-76%)
3. Antifibrotic RCT Arena
n = 24 Phase IIb/III RCTs
10 Winners vs. 14 Failures
Zero-Shot AUC = 0.87 (0.96)
4. 10x Visium Carcinoma
12 Tumor Sections
20,880 QC Spots (24k Shifts)
12/12 SEI > 0 (Z = +7.82)
5. Pre-ICB & Perturbation
n = 107 Pre-ICB + Controls
BCC 0.88 | Anti-PD-1 OS 1.95
TGFBR1i & Non-ICB Verified